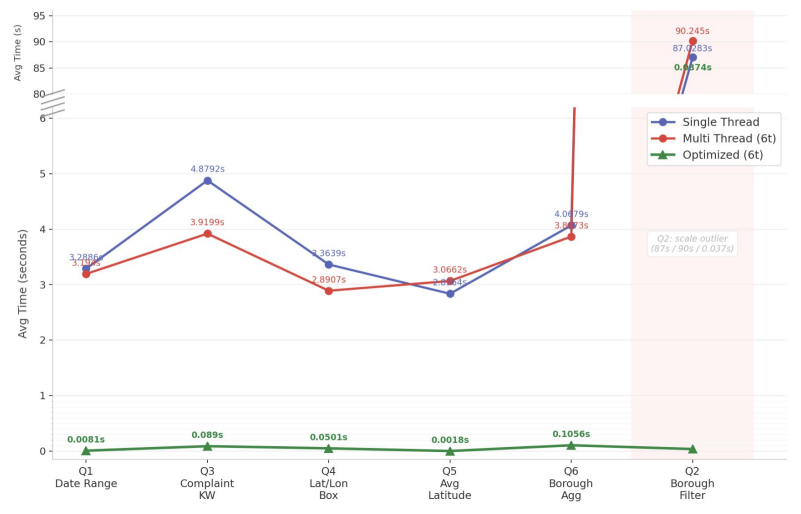
NYC 311 High-Performance Data Processing

CMPE-275 · 14M records · 6 queries · 3 Approaches · 20 runs avg

#	Query	Result Size	Single Avg (s)	Multi (6t) Avg (s)	Optimized (6t) Avg (s)
Load	Read 14M records	14,000,000	42.7317	46.2298	42.6789
1	Date range filter	484,799	3.2886	3.1940 (+2.9%)	**0.0081** (+99.8%)
2	Borough filter	4,002,729	87.0283	90.2450 (−3.7%)	**0.0374** (+99.96%)
3	Complaint keyword	165,106	4.8792	3.9199 (+19.7%)	**0.0890** (+98.2%)
4	Lat/Lon bounding box	12,673,163	3.3639	2.8907 (+14.1%)	**0.0501** (+98.5%)
5	Average latitude	scalar	2.8364	3.0662 (−8.1%)	**0.0018** (+99.9%)
6	Borough aggregation	8 zones	4.0679	3.8673 (+4.9%)	**0.1056** (+97.4%)

Performance Comparison

- Optimized code is ~72× faster on average
- Parallelism is a multiplier, not a fixer
- Naive multithreading often made queries slower



Key Discoveries

- More threads ≠ faster—not always helpful—the borough filter was **2.58× slower** with 6 threads
- Large string-heavy result sets kill parallel performance due to allocator contention
- `omp critical` in hot loops makes things **worse** as thread count grows
- Switching to **Object of Arrays** layout gave up to 99.8% speedup—a bigger win than any threading change
- Returning **pointers instead of copies** eliminated hundreds of MB of merge overhead
- Parallel CSV loading was **slower**—disk I/O can't be parallelized
- Dataset "shape" matters—strings & wide records are a bad fit for OpenMP

Memory Layout: AoS → AoA

- Array of Structs (AoS)
  Ex: Query on lat → loads all 44 fields into cache line
- Array of Arrays (OaA)
  Ex: lat: [40.7][40.8][40.6]… ← only latitudes in cache
  Query on lat → loads ONLY the latitude array

3 Key Lessons: Fix how you store data before you add threads

- Bad parallelism—the borough filter was **2.58× slower** with 6 threads. **Oops**
- Smart parallelism—average latitude was **1.36× faster** with the reduction pattern. **Better**
- The best fix wasn't threads at all—the Object **of Arrays layout gave a 99.8% speedup. Winner**

Manasa Sadhu, Bhuvana Komati, Asha Vinodhini Vijayan